

# KUNAL BAVDHANE

✉ bavdanekunal19@gmail.com

🔗 <https://github.com/kunalbavdhane9922>

🌐 [www.linkedin.com/in/kunal-bavdhane-44354032a](http://www.linkedin.com/in/kunal-bavdhane-44354032a)

## EDUCATION

### Vishwakarma Institute of Technology

Pune, Maharashtra, India

*Bachelor of Technology in Computer Engineering*  
*Current CGPA: 9.41/10.0*

June 2024 - June 2028

Relevant Coursework: Data Structures and Algorithms, Operating Systems, Database Management Systems, Object Oriented Programming.

## PROJECTS

### Sovereign AI Enterprise Protocol (SAEP) — Ongoing

- Designing a multi-tenant AI-native enterprise platform supporting AI workforce orchestration and organizational governance.
- Developing React-based frontend modules and backend APIs for workforce, company, and marketplace management.
- Designing event-driven service communication patterns using Kafka and domain-driven design principles.
- Building structured and semantic persistence layers using PostgreSQL and Qdrant for contextual knowledge retrieval.

### LeetCode Tracker & DSA Coaching Platform

JANUARY 2026

*Tools Used:* Node.js, Express.js, React, MongoDB, Redis, BullMQ, Docker

- Developed a full-stack platform that analyzes users' coding activity and generates personalized Data Structures and Algorithms (DSA) learning roadmaps based on topic-wise performance and difficulty distribution.
- Implemented asynchronous background job processing using Redis and BullMQ to periodically synchronize user problem-solving data without blocking API requests.
- Designed MongoDB aggregation pipelines to compute detailed analytics such as topic mastery, difficulty distribution, and consistency streaks, and Reduced dashboard response latency by approximately 60% using Redis caching.
- Built a spaced-repetition based revision system that automatically schedules problem reviews based on previous performance and revision intervals.

### Scalable URL Redirection Platform

FEBRUARY 2026

*Tech Used:* FastAPI, MySQL, Redis, Docker, Nginx, SQLAlchemy

- Developed a high-performance URL shortening service using Base62 encoding to generate collision-free short links stored in MySQL with optimized indexing.
- Implemented Redis cache-aside strategy to accelerate redirect operations and significantly reduce database query load for frequently accessed URLs.
- Containerized stateless FastAPI services with Docker and configured Nginx reverse proxy for load balancing to simulate horizontally scalable backend architecture.

### Real-Time Stock Market Analytics Engine

MARCH 2026

*Tech Used:* Python, FastAPI, Kafka, Redis, MySQL, Docker, React

- Built a real-time system to ingest and analyze streaming stock market data for technical pattern detection.
- Implemented financial analytics algorithms including moving averages, volume spike detection, and candlestick pattern recognition.
- Designed REST APIs using FastAPI and stored historical market data in MySQL for efficient time-series analysis.
- Optimized real-time data access using Redis caching and deployed services using Docker containers.

## RESEARCH & PUBLICATIONS

Research Papers:

1. Sign Language Recognition: A Real-Time Solution for Inclusive Communication → [Click Here](#)
2. Pattern-Based Analysis of Stock Movements Using Candlestick Charting Techniques → [Click Here](#)

## SKILLS

**Languages:** English (Fluent), Hindi (Fluent)

**Programming Languages:** C, C++, Python, Java, JavaScript, SQL

**Databases:** MongoDB, MySQL, Redis

**Frameworks & Libraries:** React, Node.js, Express, Flask, FastAPI.

**Tools & Platforms:** Docker, AWS, Git, GitHub, Nginx

**Soft Skills:** Public Speaking, Teamwork, Leadership, Communication, Management, Analytical Thinking